THE JOURNAL OF PHYSICAL CHEMISTRY B

Article

Cite This: J. Phys. Chem. B 2018, 122, 8174–8184

pubs.acs.org/JPCB

Conformational Effects in the Transport of Glucose through a Cyclic Peptide Nanotube: A Molecular Dynamics Simulation Study

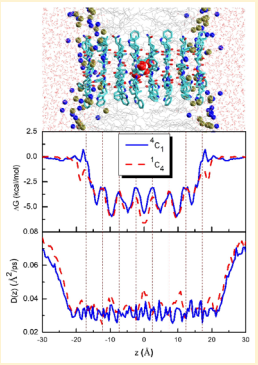
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Supporting Information

ABSTRACT: The transport behavior of glucose through a cyclic peptide nanotube (CPN), composed of 8 × cyclo[-(Trp-D-Leu)$_4$-Gln-D-Leu-] rings embedded in DMPC lipid bilayers was examined using all-atom molecular dynamics (AAMD) simulations. Two conformational isomers of β-D-glucose, equatorial ($^4C_1$) and axial ($^1C_4$) chair conformers, were used to examine conformational effects on the hydrogen bond network, energetics, and diffusivity of glucose transport through the CPN. Calculations of the number of hydrogen bonds of the two glucose conformers with water molecules and with the CPN illustrate that the total number of hydrogen bonds of the conformers decreases inside the channel compared to bulk water due to the confinement characteristics of the interior of the CPNs although new hydrogen bonds between the hydroxyl and hydroxymethyl hydrogens of glucose and the carbonyl oxygens in the CPN backbone are formed. Despite the decrease of the number of hydrogen bonds inside the CPN, intramolecular hydrogen bonds of $^1C_4$ are maintained during permeation of $^1C_4$ through the CPN. The retention of intramolecular hydrogen bonds and the spherical shape of $^1C_4$ give rise to considerably weaker orientational preferences and higher diffusion coefficients for $^1C_4$ than those of $^4C_1$ inside and outside the CPN. Due to larger dipole moments induced by the alignment of hydroxyl and hydroxymethyl groups, $^1C_4$ has more favorable interactions with the CPN backbone at the channel entrances and inside the channel than $^4C_1$. In the middle of the CPN channel, entropic gains originating from higher orientational and translational degrees of freedom of $^1C_4$ than those of $^4C_1$ also contribute to lower free energy wells for $^1C_4$ inside the CPN. This work reveals that the conformational variation and intramolecular hydrogen bond formation of β-D-glucose can have important effects on the energetics and dynamics of glucose transport through CPNs, providing insight into the translocation mechanism of D-glucose into the cell through glucose transporters (GLUTs) and the dynamics of glucose confined in silica nanochannels. It is also demonstrated that CPNs can indeed facilitate the permeation of small hydrophilic molecules such as glucose and can be utilized as a novel carrier system for hydrophilic drug compounds into the cell.



INTRODUCTION

Cell membranes act as a serious energy barrier against the transport of small hydrophilic molecules as well as molecular ions into cytoplasmic regions from extracellular regions,[1,2] and consequently, translocating hydrophilic molecules such as glucose into cells is mediated by channel or transporter proteins.[3−5] Recently, synthetic channel systems that span through the cell membrane have emerged as important structures for delivering hydrophilic molecules as well as molecular ions into the cytoplasmic region.[6−8]

Cyclic peptide nanotubes (CPNs), which are a family of synthetic ion channels, involve the formation of tubular structures through intermolecular hydrogen bonding between adjacent cyclic peptide rings composed of alternating D- and L-type amino acid residues.[9−21] The diameter of CPNs can be modulated by altering the number of amino acid residues that comprise a cyclic peptide ring, and as a result, CPNs with larger diameters can translocate hydrophilic small molecules or molecular ions across cell membranes due to the hydrophilic nature of the interior of CPNs. Unlike natural channel proteins that selectively and actively permeate ions or small biomoles across cell membranes, CPNs passively transport such ions or molecules due to lack of active functionality in their simple tubular strucure and thus have limitations in the applications. Nevetheless, a number of experimental and computational studies have shown potential applications of CPNs in several research areas such as biomedical science.[15−25] Synthesizing CPNs formed by cyclic decapeptide rings, Granja and co-workers showed that glucose, a hydrophilic biomolecule, can permeate through CPNs embedded in lipid bilayers for the first time.[22] Sánchez-Quesada et al. reported that unlike cyclic octapeptide nanotubes that lack glutamate transport activity, cyclic decapeptide nanotubes can allow the transport of glutamate ions from the inside of vesicles to the extravesicular region by measuring the glutamate transport rates across vesicular membranes.[24] It has recently been demonstrated both

Received: June 12, 2018
Revised: August 3, 2018
Published: August 7, 2018

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DOI: 10.1021/acs.jpcb.8b05591
J. Phys. Chem. B 2018, 122, 8174–8184

experimentally and computationally that CPNs can be used as a potential drug delivery system through observations of the release of 5-fluorouracil, an anticancer drug, from liposomes through a cyclic decapeptide nanotube.[15,17,21] In addition, the transport behavior of simple organic molecules including ethanol and lactic acid through CPNs was studied using molecular dynamics (MD) simulations.[19,20,26,27] D-Glucose (also known as D-glucopyranose), which is one of vital hydrophilic biomolecules, plays an important role in cell metabolism.[28] D-Glucose is the building block of polysaccharides such as starch and cellulose and acts as an energy source for most prokaryotic and eukaryotic cells. Many experimental and computational studies have been devoted to understanding of the uptake mechanism of D-glucose into the cell through glucose transporters (GLUTs).[29−32] D-Glucose is also known to protect and maintain the protein and membrane structure in a living cell under an extreme condition such as freezing or drying, and there have been many studies of the structural and dynamic behavior of glucose inside hydrophilic silica nanochannels that mimic biologically confined interfaces like protein or membrane surfaces.[33−35] β-D-Glucose, which is one of the typical conformational isomers of D-glucose, can take two chair conformations. In the equatorial chair form of β-D-glucose designated as $^4C_1$, all the hydroxyl and hydroxymethyl groups are placed in equatorial positions (Figure 1a).

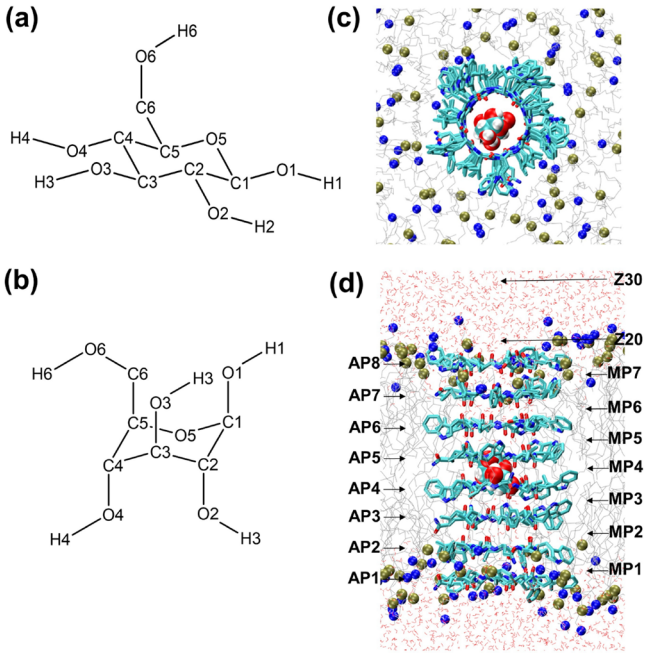


Figure 1. (a) Equatorial ($^4C_1$) and (b) axial ($^1C_4$) conformational isomer structures of the β-D-glucose and (c) top and (d) side views of a cyclic peptide nanotube (CPN), 8 × cyclo[-(Trp-D-Leu)$_4$-Gln-D-Leu-] with a $^1C_4$ β-D-glucose molecule inside in DMPC lipid bilayers. In (c) and (d), hydrogen atoms of the CPN are removed for clarity, lipid molecules are highlighted by transparent gray tubes, and P and N atoms in the lipid head groups are represented by tan and blue spheres, respectively. In (d), the α-plane (AP) regions and midplane (MP) regions are defined in the text following the work in ref 11. The AP regions in the z direction are positioned at ±2.45, ±7.35, ±12.25, and ±17.15 and the MP regions are positioned between the two adjacent AP regions. The channel entrances of the CPN are located near z = ±20.0 Å (Z20) and Z30 is defined as the position of the glucose at z = 30 Å.

However, an alternative chair form, designated as $^1C_4$, takes all the ring substituents in axial positions (Figure 1b). The syn-diaxial positioning of the hydroxyl and hydroxymethyl groups in the $^1C_4$ conformation leads to strong intramolecular hydrogen bonds between the hydroxyl and hydroxyl or hydroxymethyl groups. It has been suggested that intramolecular hydrogen bond cooperativity of carbohydrates may play an important role in protein−carbohydrate recognition and the cryoprotection of carbohydrates.[36] In order to determine accurate energy difference between the $^4C_1$ and $^1C_4$ conformations, many experimental and computational studies have been conducted and have revealed that due to considerable free energy gaps between the two conformations greater than about 5 kcal/mol, most β-D-glucose molecules have a conformational preference for the $^4C_1$ form in water.[37−44]

In this study, we investigate the transport behavior of β-D-glucose through a CPN embedded in a DMPC lipid bilayer. We use the all-atom MD (AAMD) simulation method to describe the structural and dynamic properties of β-D-glucose in a confined region such as a CPN at the atomistic level. Because CPNs share the hydrophilic nature of their interior surfaces with silica nanochannels, we believe that our study also sheds light on the dynamics of glucose confined in silica nanochannels as well as the uptake mechanism of D-glucose into the intracellular region through GLUTs. Although the $^4C_1$ conformation is the preferential form found in β-D-glucose, it is of interest to realize the effect of the conformational difference on the transport behavior of glucose through CPNs, and as a result, the $^1C_4$ form is also taken into account in this work. We calculate orientational order parameters and the number of hydrogen bonds of the glucose conformers with water molecules and with the CPN backbone to examine conformational effects on the orientational preference and hydrogen bonding network of glucose inside CPNs. In order to assess the energetics and dynamics of glucose transport through CPNs, we obtain the potential of mean force (PMF) and position-dependent diffusion coefficient profiles for the two chair conformers of β-D-glucose using the adaptive biasing force (ABF) and Bayesian inference (BI) methods combined with MD simulation trajectories.[18,45]

The paper is organized as follows. The simulation systems and methods are introduced in next section. The calculated results are provided in the Results and Discussion, and conclusions and further work are addressed in the final section.

COMPUTATIONAL METHODS

Description of the Simulations. The model systems are composed of a single CPN, 80 DMPC (1,2-dimyristoyl-*sn*-glycero-3-phosphocholine) lipid molecules, 3808 water molecules, and a single equatorial ($^4C_1$) or axial ($^1C_4$) β-D-glucose molecule in a simulation box the dimensions of which are approximately 53 Å × 50 Å × 81 Å (Figure 1c,d). Because decapeptide CPNs have larger diameters than octapeptide CPNs, they allow small molecules such as glucose to permeate through themselves.[15,22,46] The decapeptide CPN used in this work is composed of 8 × cyclo[-(Trp-D-Leu)$_4$-Gln-D-Leu-] peptide rings, which can be classified into three types with respect to the relative position of the Gln residue in peptide rings.[15] We chose a CPN with the Gln residue in the meta position because MD simulations that we had performed separately showed this type of a CPN to be most stable inside DMPC lipid bilayers. The channel axis of the CPN was aligned

DOI: 10.1021/acs.jpcb.8b05591
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parallel to the z axis of the system that is normal to the DMPC lipid bilayer surface. All the simulations were carried out in the NpT ensemble with periodic boundary conditions applied in all directions. Simulation systems were fully equilibriated at 303 K and 1 bar using the Langevin thermostat and the Nosé–Hoover Langevin piston method. The damping coefficient, γ_L in the Langevin thermostat considerably affects the diffusion coefficient of a molecule in the simulation box. A damping coefficient, γ_L = 5.0 ps^{-1} was selected in our MD simulations because the mean-square displacement (MSD) analysis combined with an MD simulation for $^4C_1$ in bulk water with that value of γ_L generated a diffusion coefficient of 0.067 Å^2/ps that was close to an experimental value of 0.069 Å^2/ps.[47,48] The electrostatic potential energy was calculated using the particle mesh Ewald (PME) method with conducting boundary conditions. The van der Waals interactions were truncated at 12 Å and were smoothly switched to 0 from 10 to 12 Å. The CHARMM22 and CHARMM36 force field parameter sets were used for the CPN and DMPC lipid molecules, respectively, and the TIP3P water model for water. The force field parameters for the β-D-glucose conformers were obtained from the work by Kuttel and co-workers.[49] All the MD simulations were performed using NAMD 2.8 and visualizations were done using VMD 1.8.[50,51]

Construction of PMF and Position-Dependent Diffusion Coefficient Profiles for the β-D-Glucose Conformers. Symmetrized PMF profiles for the two glucose conformers of β-D-glucose were calculated as a function of the z coordinate of the center of mass (CoM) of each conformer using adaptive biasing force (ABF) and thermodynamic integration (TI) methods. For PMF calculations, the channel axis of the CPN was kept parallel to the z direction using the colvar method. The CoM of the C_α atoms in each peptide ring was restrained in the corresponding α-plane regions with a spring constant of 25 (kcal/mol)/Å^2 to keep the distance between the two adjacent peptide rings to be 4.9 Å in the z direction (see Figure 1d). To avoid a divergence problem due to integration over all xy areas outside the CPN, the CoM of each glucose molecule was kept within 2.5 Å in the xy plane from the channel axis by applying a weak harmonic restraint with a spring constant, k_{xy}^{glu}, of 0.16 (kcal/mol)/Å^2 to the glucose. The reaction coordinates of the glucose in the z direction spanning from −30.0 to +30.0 Å were divided into six sections, and each sectional PMF profile was evaluated using the ABF method. The bin width in the sectional PMF profiles was set to be 0.4 Å. Respective contributions from the CPN, water, and DMPC bilayers to the energetics of glucose transport through the CPN were examined by decomposing the total symmetrized PMF profiles for the glucose conformers via the TI method.[52] In this method, the mean forces acting on the glucose from the CPN, water, and DMPC bilayers were separately computed and integrated from the corresponding ABF MD trajectories. The statistical errors in the PMF and PMF decomposition profiles were estimated using the method proposed in ref 53.

To assess the dynamic behavior of glucose inside the channel, position-dependent diffusion coefficient calculations of the glucose conformers along the channel axis (z axis), $D(z)$ were performed using a Baysian inference (BI) method combined with a Monte Carlo (MC) technique.[18,45] The observation time or lag time, Δt, required in the BI/MC method affects the diffusivity of a molecule or an ion under a non-Markovian condition such as subdiffusion.[18,52,53] In our study, the dependence of diffusivity of glucose on Δt was investigated by calculating the position-dependent diffusion coefficients with three different values of Δt, namely, 1.2, 2.0, and 2.8 ps. Figure S1 in the Supporting Information reveals the dependence of the position-dependent diffusion coefficients of the glucose conformers on Δt, indicating that the diffusivity of the glucose molecules inside the CPN is indeed a non-Markovian process. In our calculations Δt was set to be 2.0 ps because the position-dependent diffusion coefficients of the glucose conformers obtained by the BI/MC method with that value of Δt agree well with those obtained from the MSD method inside the CPN. The ABF trajectories used for the calculation of the PMF profiles were employed as input coordinates for the BI/MC method where a position-dependent diffusion coefficient profile for a glucose conformer was calculated as a function of the z coordinate of the CoM of the glucose conformer.

Definition of a Hydrogen Bond. Intermolecular or intramolecular hydrogen bonds of a glucose molecule with the CPN backbone, water molecules, and glucose itself were analyzed using the definition in ref 54, where a hydrogen bond is considered to be formed between two molecules or inside a molecule if their interoxygen distance, R_{OO} < 3.5 Å and simultaneously, the angle between the O–O axis and the O–H bond, θ_{HOO} < 30°.

RESULTS AND DISCUSSION

Figure 2 shows the number of hydrogen bonds (N_{hb}) that a glucose conformer forms with water molecules, the carbonyl oxygen atoms, and amide hydrogens of the CPN backbone, and itself during the permeation through the CPN using the hydrogen bond definition introduced in the previous section. While negligible intramolecular hydrogen bonds are formed in

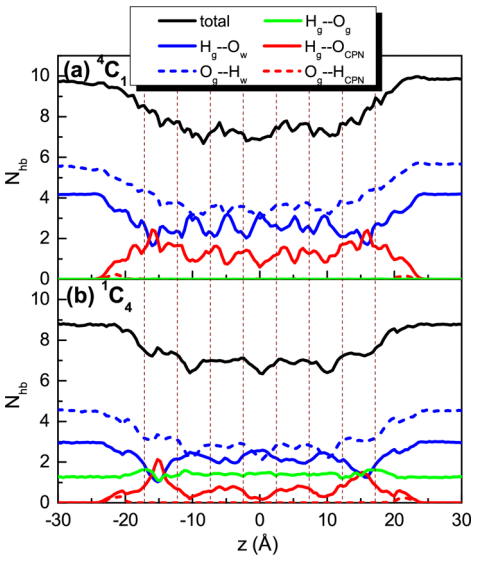


Figure 2. Number of hydrogen bonds (N_{hb}) of (a) $^4C_1$ and (b) $^1C_4$ with water molecules, the channel backbone, and itself during the permeation through the 8 × cyclo[-(Trp-D-Leu)$_4$-Gln-D-Leu-] in DMPC bilayers. O_g and H_g in the legend represent the hydroxyl oxygen and hydrogen of β-D-glucose, O_w and H_w represent the water oxygen and hydrogen, and O_{CPN} and H_{CPN} represent the carbonyl oxygen and amide hydrogen in the backbone of the CPN. The vertical broken lines indicate the position of each cyclic peptide rings in the z direction.

DOI: 10.1021/acs.jpcb.8b05591
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$^4C_1$ in Figure 2a, a number of intramolecular hydrogen bonds larger than one are observed to be formed in $^1C_4$ and they are maintained during the permeation of $^1C_4$ through the CPN in Figure 2b. It has already been reported in a previous study that most intramolecular hydrogen bonds are formed only in $^1C_4$ due to structural distinctions between $^4C_1$ and $^1C_4$ in Figure 1a,b.[36] As the glucose conformers approach the CPN, there is a reduction of N_{hb} that begins outside the channel entrances. This is followed by an increase in N_{hb} for the hydroxyl hydrogens of each glucose conformer involving carbonyl oxygens in the CPN backbone, implying that the reduction of N_{hb} between each glucose conformer and water molecules at the channel entrances is associated with dangling carbonyl oxygens in the outermost peptide rings that compete against water molecules for hydrogen bond formation. Comparison of Figure 2a,b reveals that the total number of hydrogen bonds and the number of intermolecular hydrogen bonds with the CPN and water molecules in $^1C_4$ are smaller than in $^4C_1$ due to reduction in intermolecular hydrogen bonds resulting from the intramolecular hydrogen bonds formed only in $^1C_4$. It is observed that the hydroxyl groups of both glucose conformers predominantly act as hydrogen bond donors to the CPN because the shorter N—H bonds (compared to the carbonyl C═O bonds in the CPN backbone) are incapable of accessing the hydroxyl oxygen atoms of the glucose conformers and fail to form hydrogen bonds with them. It it worth noting that the reduction in N_{hb} of the glucose conformers as a hydrogen bond donor with water molecules are nearly canceled out by the increase in N_{hb} of the hydroxyl hydrogens of the glucose conformers with the carbonyl oxygens of the CPN, indicating that there is competition between the carbonyl oxygens of the CPN backbone and the water oxygens for hydrogen bond formation with the hydroxyl groups of the glucose conformers. As a result, the net reduction of the total number of hydrogen bonds inside the CPN is mainly due to the fact that the reduction of the hydrogen bonds of the hydroxyl oxygens of the glucose with water hydrogens is not compensated for by the negligible number of hydrogen bonds with the hydroxyl oxygens of the glucose with the amide hydrogens of the CPN backbone.

To further assess the hydrogen bond network that a glucose conformer develops during permeation across the CPN, the number of hydrogen bonds (N_{hb}) that each hydrogen and oxygen atom in the hydroxyl groups of the glucose conformers form with water molecules, the carbonyl oxygens in the CPN backbone, and themselves are illustrated in Figure 3. In Figure 3a–e, N_{hb} values of each hydroxyl hydrogen and oxygen atom of $^4C_1$ with water molecules at the $z = \pm 30$ Å agree well with those in bulk water that were reported by previous studies although a hydrogen bond was defined in a slightly different manner.[43,55] In the $^4C_1$ conformer, N_{hb} between H6 and O_w has a strong negative correlation with that between H3 and O_w and has a modest positive correlation with that between H1 and O_w (Figure 3a,c,e). The H1 and H6 atoms of $^4C_1$ have more hydrogen bonds with water oxygens when the CoM of $^4C_1$ is in the midplane regions than when it is in the α-plane regions while the H3 of $^4C_1$ has more hydrogen bonds with water oxygens when the CoM of $^4C_1$ is in the α-planes. In a previous study, interactions between a cation and water oxygens were shown to be more favorable when a cation is located in the α-plane regions due to the lack of competition between the carbonyl oxygens and water oxygens for a cation at that location,[56] and this applies to the hydrogen bonding

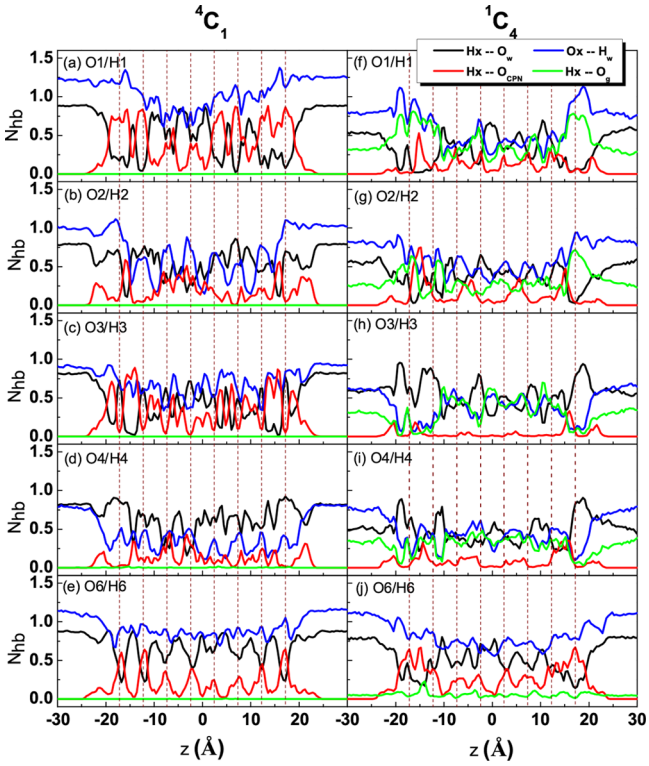


Figure 3. Numbers of hydrogen bonds (N_{hb}) of each hydrogen and oxygen atom in the hydroxyl groups of (a)–(e) $^4C_1$ and (f)–(j) $^1C_4$ with water molecules, carbonyl oxygens of the CPN backbone, and themselves during the permeation through the CPN. In the legend, x indicates the atom numbering of $^4C_1$ and $^1C_4$ in Figure 1a,b. O_w, H_w, O_{CPN}, and O_g represent the oxygen and hydrogen atom of the water molecule, the carbonyl oxygen in the backbone of the CPN, and the oxygen atom of the glucose, respectively. The vertical broken lines are the same as those in Figure 2.

between a hydroxyl hydrogen of the glucose and the water oxygen. When the CoM of $^4C_1$ resides in a midplane (α-plane) region, the H1 and H6 of $^4C_1$ are positioned in adjacent α-plane (midplane) regions and form more hydrogen bonds with water oxygens than with the carbonyl oxygens in the CPN backbone, but the H3 hydrogen is located in the same midplane (α-plane) region and has more chances to form a hydrogen bond with the carbonyl oxygens in the CPN backbone. In the case of $^1C_4$, an increase in intramolecular hydrogen bonds for H1 and H2 at the channel entrance is observed in Figure 3f,g. While intramolecular hydrogen bonds for H3 and H4 decrease, the N_{hb} values of H3 and H4 with water oxygens increase at the channel entrances in Figure 3h,i. Because H1 of $^1C_4$ can make an intramolecular hydrogen bond only with O3 or O6 and H2 can do only with O4, the increase in intramolecular hydrogen bonds for H1 or H2 will lead to a reduced number of intramolecular hydrogen bonds for H3 or H4. Instead, H3 and H4 will have more chance to form a hydrogen bond with water oxygens. Compared to the case for $^4C_1$, hydrogen bond formation of the hydroxyl hydrogens of $^1C_4$ with carbonyl oxygens in the CPN backbone except for H6 of the hydroxymethyl group is suppressed owing to the preservation of intramolecular hydrogen bonds of $^1C_4$ inside the channel.

Configurations that a glucose conformer takes inside a CPN are examined by calculating the deviations of the z coordinate of the hydroxyl, hydroxymethyl and endocyclic oxygen atoms

DOI: 10.1021/acs.jpcb.8b05591
J. Phys. Chem. B 2018, 122, 8174–8184

from the z coordinate of the CoM of a glucose conformer. Here Δz_O is defined as $\Delta z_O = z_O - z_{CoM}$ where z_O is the z coordinate of the hydroxyl, hydroxymethyl or endocyclic oxygen atom and z_{CoM} is the z coordinate of the CoM of a glucose conformer. The evaluated Δz_O profiles in Figure 4 show larger deviations in $^4C_1$ than in $^1C_4$ due to difference in the positioning of the hydroxyl and hydroxymethyl groups in the two glucose conformers. Figure 4a for $^4C_1$ shows strong positive correlations between Δz_{O1} and Δz_{O2} and between Δz_{O4} and Δz_{O6}, especially, at the channel entrances. Taking into account that the values of Δz_{O3} and Δz_{O5} are relatively small and negatively correlated, large positive values of Δz_{O1} and Δz_{O2} and large negative ones for Δz_{O4} and Δz_{O6} at the upper channel entrance near the Z20 region indicate that when $^4C_1$ is entering into the channel of the CPN, the O4 and O6 atoms are directing inward and the O1 and O2 are pointing outside while O3 and O5 are mostly facing the opposite channel walls rather than orienting outward or inward. The Δz_O values for $^1C_4$ show similar trends, but they are less pronounced and less correlated than those of $^4C_1$, indicating that $^1C_4$, which is more spherical than $^4C_1$ and can form intramolecular hydrogen bonds, has more rotational degrees of freedom inside the channel and fewer orientational preferences than $^4C_1$. In Figure 4b, the oscillating behavior of the Δz_{O6} of $^1C_4$ inside the channel demonstrate that $^1C_4$ can rotate or change directions in the midplane or α-plane regions periodically.

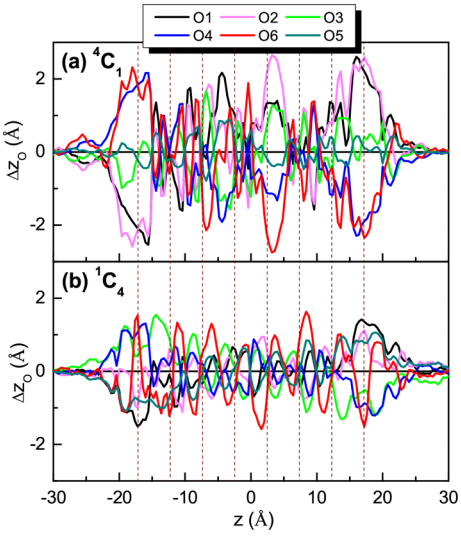


Figure 4. Deviations of the z_O coordinate of some oxygen atoms of (a) $^4C_1$ and (b) $^1C_4$ from the CoM of each glucose conformer where Δz is defined as $\Delta z = z_O - z_{CoM}$. The horizontal black line in the middle in the graphs indicates the position of z_{CoM}. The vertical broken lines are the same as those in Figure 2.

The extent to which a glucose molecule forms hydrogen bonds with water or the carbonyl oxygens of the CPN can be analyzed by calculating radial distribution functions (RDF) for the hydroxyl hydrogens of the glucose. In Figure 5a–d, as the maximum peaks of the RDFs of the hydroxyl hydrogens of $^4C_1$ with the water oxygens increase, the maxima of the RDFs with the carbonyl oxygens of the CPN backbone decrease due to competition between the water oxygen and the carbonyl

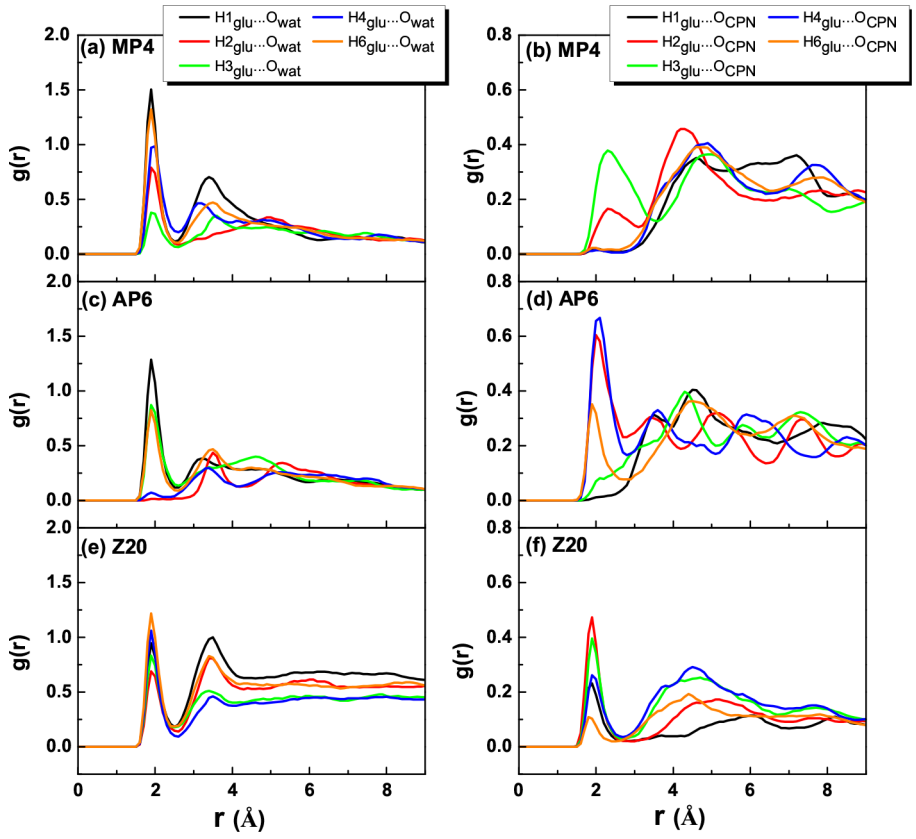


Figure 5. Radial distribution functions (RDFs) for the hydroxyl hydrogens of $^4C_1$ with water oxygen atoms and the carbonyl oxygen atoms of the backbone of the CPN in the MP4, AP6, and Z20 regions.

DOI: 10.1021/acs.jpcb.8b05591
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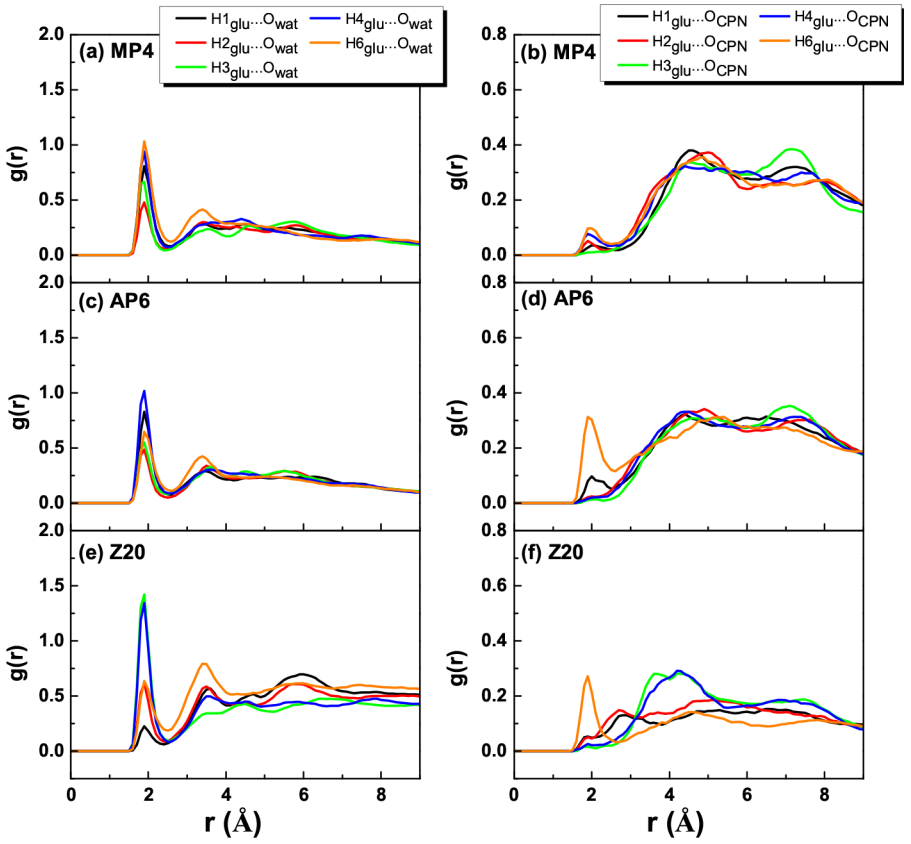


Figure 6. Radial distribution functions (RDFs) for the hydroxyl hydrogens of $^1C_4$ with water oxygen atoms and the carbonyl oxygen atoms of the backbone of the CPN in the MP4, AP6, and Z20 regions.

oxygen atoms for the formation of hydrogen bonds with the hydroxyl hydrogens of the glucose mentioned in Figure 2a. Parts a and c of Figure 5 show that the hydroxyl hydrogens of $^4C_1$ except for the H3 hydrogen have more hydrogen bonds with water oxygens in the midplane regions than in the α-plane regions of the CPN, indicating that when $^4C_1$ resides in a midplane region, the hydroxyl hydrogens of $^4C_1$ except for H3 are placed near the adjacent α-plane regions which molecules can easily access but the carbonyl oxygen atoms of the CPN backbone can hardly reach. In Figure 5b,d, the hydroxyl hydrogen atoms of $^4C_1$ take more hydrogen bonds with the carbonyl oxygens of the backbone when $^4C_1$ are placed in the α-plane regions where the hydroxyl hydrogens have more chances to contact the carbonyl oxygens, which also agrees well with the results from Figure 2a. The RDF profiles for the hydroxyl hydrogens of $^4C_1$ demonstrate that the pyranose plane of $^4C_1$ is likely to stay parallel to the channel axis (z axis) such that the hydroxyl hydrogens of $^4C_1$ can access the α-plane regions when the CoM of $^4C_1$ is placed in the midplane regions and vice versa. In the Z20 region, which is close to one of the channel entrances, the first maximum of the RDF for H6 with water oxygens is higher than those of H1 and H2 (Figure 5e) while the first maxima of the RDFs for H1 and H2 with the carbonyl oxygens of the CPN is much higher than that of H6 (Figure 5f). Taking into account that $\Delta z_{O1} > 0$, $\Delta z_{O2} > 0$ and $\Delta z_{O6} < 0$ near the Z20 region in Figure 4a, we can speculate that the O6−H6 group of $^4C_1$ is oriented toward the interior of the channel such that H6 can interact with water molecules around the AP8 region while the O1−H1 and O2−H2 groups are directing outside the channel where H1 can interact with both the water oxygens outside the channel and the dangling carbonyl oxygens in the uppermost cyclic peptide ring.

Parts a−d of Figure 6 show that the peak maxima in the RDFs for the hydroxyl hydrogens of $^1C_4$ with water molecules and with the carbonyl oxygens of the CPN are lower than those of $^4C_1$ and are not well-defined in the RDFs for the hydroxyl hydrogens with the carbonyl oxygens owing to the intramolecular hydrogen bonds of $^1C_4$ maintained inside the channel as noted in Figure 2b. Combined with the fact that $\Delta z_{O3} < 0$ and $\Delta z_{O4} < 0$ in Figure 4b are near the Z20 region, much higher peak maxima in the RDFs for the H3 and H4 hydrogens than the other hydroxyl hydrogens in Figure 6e demonstrate that the O3−H3 and O4−H4 hydroxyl groups of $^1C_4$ are oriented toward the interior of the CPN at the channel entrances and interact strongly with the water oxygens residing in the AP8 region. Figure 6f reveals that in contrast with $^4C_1$, the other hydroxyl hydrogens of $^1C_4$ except for the H6 hydrogen weakly interact with the dangling carbonyl oxygens at the uppermost cyclic peptide rings due to intramolecular hydrogen bonds, which is also consistent with the results in Figure 3f−j.

For a closer look into orientational behaviors of the glucose conformers inside the CPN, orientational order parameters, $S(z)$, for both conformers were calculated where $S(z)$ is defined as

$$S(z) = \frac{1}{2}\langle 3\cos^2\theta - 1\rangle_z \quad (1)$$

Here θ is the angle between the z axis and the normal vector to the plane generated by C1, C2, and C5 of the glucose molecule (see Figure 1a,b) and z is z coordinate of the CoM of a glucose

DOI: 10.1021/acs.jpcb.8b05591
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conformer. In Figure 7, the $S(z)$ values of $^4C_1$ close to −0.5 in the midplane regions indicate that the normal vector to the

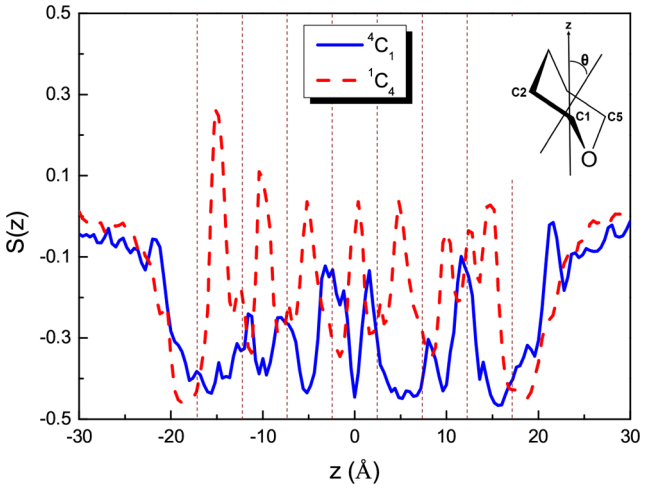


Figure 7. Orientational order parameters of (a) $^4C_1$ and (b) $^1C_4$ through the CPN. The definition of the angle between the normal to the C1−C2−C5 plane of the glucose and the channel axis (z axis) is provided in the inset. The vertical broken lines are the same as those in Figure 2.

C1−C2−C5 plane of $^4C_1$ tends to be perpendicular to the z axis at a midplane region. Namely, the C1−C2−C5 plane of $^4C_1$ stays parallel to the z axis. As mentioned above, the hydroxyl groups of $^4C_1$ except for the O3−H3 group are confined in in the adjacent α-plane regions when the CoM of $^4C_1$ is in a midplane region, and to keep hydrogen bonds with water molecules in a narrow space like the α-plane regions of the CPN, only the parallel configurations of the C1−C2−C5 plane of $^4C_1$ to the z axis are allowed. In the α-plane regions, however, $^4C_1$ shows mild orientational preferences because the hydroxyl groups are placed at the larger midplane regions and need to rotate to interact with the carbonyl oxygens close to the channel walls. The $^1C_4$ conformer, which has a more spherical shape than $^4C_1$, can rotate more freely in the spacious midplane regions and shows very weak orientational preferences as indicated by $S(z)$ close to zero in the midplane regions. Near the channel entrances, the $S(z)$ values of both conformers are close to −0.5, indicating that the C1−C2−C5 plane tends to be parallel to the channel axis to avoid the steric obstacles for the entry into the channel.

Typical configurations of the two glucose conformers along with water molecules inside the CPN are illustrated in Figure 8. Parts a and b of Figure 8 indicate that with the hydroxymethyl group pointing toward the interior of the CPN, the hydroxyl groups of $^4C_1$ play as hydrogen bond donors to the CPN, forming one or two hydrogen bonds with the carbonyl oxygens of the CPN backbone in the AP6 and MP4 regions. To water molecules, the hydroxyl groups act as both hydrogen bond donors and acceptors, which agrees well with the results in Figure 2a. It is also seen in Figure 8a that one water oxygen can form two hydrogen bonds with two different hydroxyl hydrogens of $^4C_1$ inside the channel. As noticed in Figure 2b, the numbers of hydrogen bonds of $^1C_4$ with the CPN backbone and water molecules are reduced due to intramolecular hydrogen bonds formed inside $^1C_4$. The intramolecular hydrogen bonds between O1 and H3 and between O4 and H2 are clearly observed in Figure 8c,d,

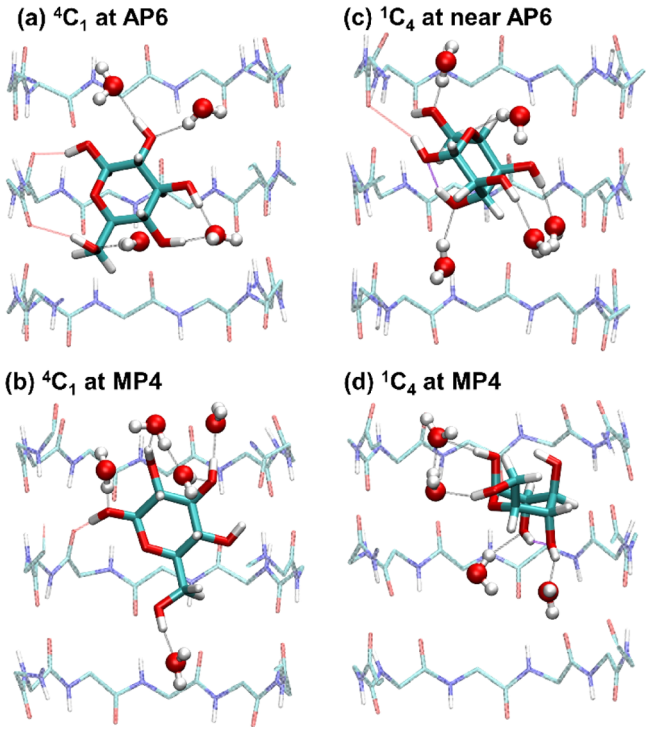


Figure 8. Typical configurations of $^4C_1$ and $^1C_4$ interacting with the CPN backbones and water molecules in the AP6 (or near AP6) and MP4 regions. (a) and (b) for $^4C_1$ and (c) and (d) for $^1C_4$. Following the same definition of the hydrogen bonding as in the Computational Methods section, a hydrogen bond between a donor and an acceptor is represented with the broken lines. Side chains of the CPN were removed for clarity, and only the water molecules forming hydrogen bonds with the glucose conformers inside the CPN are shown.

respectively. Hydrogen bonds of the hydroxyl groups of $^1C_4$ with water molecules and with the carbonyl oxygen atoms of the CPN backbone are also shown in the near AP6 and the MP4 regions in Figure 8c,d.

The symmetrized PMF profiles for $^4C_1$ and $^1C_4$ through the CPN in Figure 9 feature global energy minima lower than −5 kcal/mol, clearly demonstrating that CPNs can indeed accommodate biomolecules such as glucose and can be used as a drug delivery system.[15] It is also noticed in the PMF

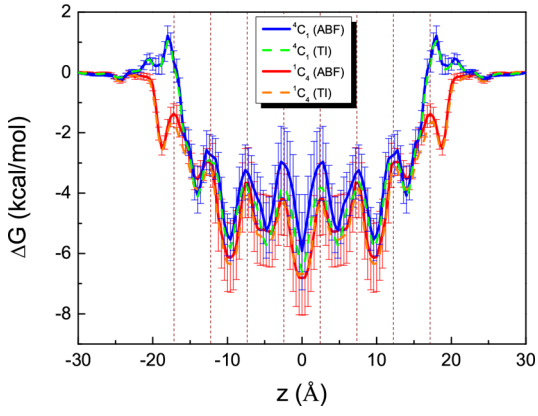


Figure 9. Symmetrized PMF profiles of $^4C_1$ and $^1C_4$ conformers through the 8 × cyclo[-(Trp-D-Leu)$_4$-Gln-D-Leu-] in DMPC bilayers obtained from the ABF and TI methods. The vertical broken lines are the same as those in Figure 2.

DOI: 10.1021/acs.jpcb.8b05591
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profiles that local minima are placed in the midplane regions and local barriers in the α-plane regions possibly due to better hydrogen bond formations with water molecules and favorable interactions with the CPN at the midplanes for the glucose conformers. The considerably weaker orientational preferences of $^1C_4$ than those of $^4C_1$ shown in Figure 8 imply that configurational entropy due to larger rotational degrees of freedom can also contribute to deeper energy wells of $^1C_4$ inside the channel, and the PMF for $^1C_4$ displays slightly deeper energy wells than $^4C_1$ in most regions through the CPN. A sharp difference between the two conformers in the PMF profiles appears at the channel entrances where $^4C_1$ has energy barriers while the $^1C_4$ has energy wells. When a hydrophilic glucose molecule enters into a confined channel entrance, several water molecules around the glucose are removed and dehydration energy penalties for the glucose arise. As a result, the energy barriers at the channel entrances for $^4C_1$ can be explained by dehydration energy penalties that are not compensated for by stabilization energies from interactions with the CPN. On the contrary, it is inferred that stabilization energies from interactions of $^1C_4$ with the CPN are large enough to cancel out dehydration energies, leading to energy wells at the channel entrances.

The PMF decomposition profiles in Figure 10 illustrate that as noted in the PMF profiles, the $^1C_4$ conformer has stronger

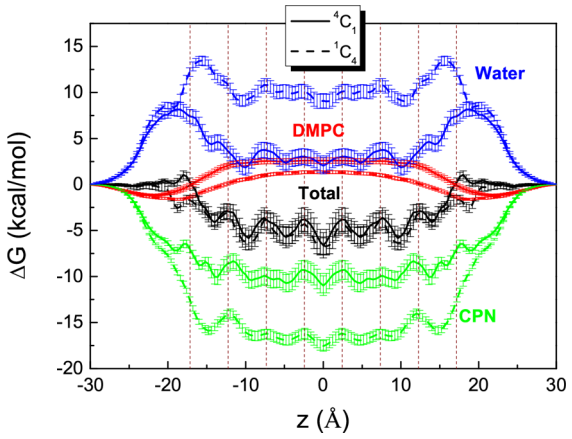


Figure 10. PMF decomposition profiles of the $^4C_1$ and $^1C_4$ conformers of β-D-glucose through the 8 × cyclo[-(Trp-D-Leu)$_4$-Gln-D-Leu-] in DMPC bilayers obtained from the TI method. The vertical broken lines are the same as those in Figure 2.

interaction energies with the CPN than $^4C_1$ at the channel entrances as well as inside the channel. This compensates for the dehydration penalties for $^1C_4$, while the dehydration energy penalties of $^4C_1$ at the channel entrances are not overcome by interactions with the CPN. In Figure 10 high dehydration penalties for $^4C_1$ near the channel entrances decrease gradually as $^4C_1$ enters into the interior of the CPN and dehydration penalties become marginal near the middle of the channel, indicating that favorable interactions of $^4C_1$ with the channel waters are recovered inside the channel. However, the resilience of interactions between $^4C_1$ and water molecules inside the channel reduces interactions of $^4C_1$ with the CPN. As a result, $^4C_1$ has shallower energy wells in the PMF than $^1C_4$. In contrast, $^1C_4$, which possesses intramolecular hydrogen bonds, has fewer intermolecular hydrogen bonds and more unfavorable interactions with water molecules inside the CPN than $^4C_1$, leading to more favorable interactions of $^1C_4$ with the CPN. The PMF and PMF decomposition profiles for the glucose conformers inside the CPN demonstrate that molecular structure and intramolecular hydrogen bond activity of glucose can deeply affect protein−glucose recognition through variations in the protein−glucose interaction.[36]

The hydroxyl and hydroxymethyl groups of glucose play a crucial role in the electrostatic interaction as well as the hydrogen bond formation of glucose with other molecules. To examine the role of the hydroxyl and hydroxymethyl groups in electrostatic interactions between the glucose conformers and the CPN, the magnitudes of the dipole moment and the tilt angles of the dipole moment and the hydroxyl and hydroxymethyl groups with respect to the channel axis for the glucose conformers are presented in Figure 11. The magnitudes of the dipole moment for $^4C_1$ at $z = \pm 30.0$ Å in Figure 11a are close to the values for several glucose conformers in the solution phase obtained from density functional theory calculations.[57] In Figure 11a,b, in contrast to $\langle\mu_x\rangle$ and $\langle\mu_y\rangle$ that are averaged out through the channel, the $\langle\mu_z\rangle$ values of the glucose conformers uniformly vary along the channel axis and the dipole moments of the glucose conformers near the channel entrances are aligned along the channel axis directing toward the interior of the channel. This demonstrates that there are orientational preferences of the glucose dipoles along the channel axis due to electrostatic potentials generated by the atomic charge distributions of the CPN backbone.[13] Such orientational behavior of the dipole moment of the glucose conformers was also observed in the behavior of water dipole moments inside CPNs.[18,58] Parts c and d of Figure 11 reveal that the increase of $|\mu|$ of the glucose molecules at the channel entrances observed in Figure 11a,b are strongly associated with the orientational rearrangements of the hydroxyl and hydroxymethyl groups of the glucose conformers at the channel entrances and that there are strong orientational correlations between $\langle\mu_z\rangle$ and the hydroxyl and hydroxymethyl groups. It is worth noting that the orientational correlations are more enhanced in $^1C_4$ rather than in $^4C_1$. Consequently, the much lower free energy wells of $^1C_4$ with the CPN than $^4C_1$ in the PMF decomposition profiles shown in Figure 10 can be attributed to more favorable electrostatic interactions of the dipole moment of $^1C_4$ with the CPN backbone.

Finally, position-dependent diffusion coefficients of $^4C_1$ and $^1C_4$ obtained from the BI/MC method with $\Delta t = 2.0$ ps are presented in Figure 12. Figure 12 shows that the position-dependent diffusion coefficients from the BI/MC method are in good agreement with those obtained from the MSD method in several positions inside the CPN, confirming a proper choice of $\Delta t = 2.0$ ps. Unlike the PMF profiles in Figure 9, the position-dependent diffusion coefficients of both glucose conformers decrease more than 10 Å away from the channel entrance, and end up being 40−50% of the coefficients in bulk water inside the channel. This kind of behavior of the position-dependent diffusion coefficients of the glucose conformers has also been observed for cations diffusing through CPNs.[52] The calculated diffusion coefficients of $^1C_4$ in the exterior and interior of the CPN are generally larger than those of $^4C_1$ possibly because intramolecular hydrogen bonds in $^1C_4$ reduce the formation of intermolecular hydrogen bonds with water molecules, resulting in smaller hydrodynamic radii of $^1C_4$ than those of $^4C_1$. Inside the CPN the diffusion coefficient profile for $^4C_1$ shows mild fluctuations inside the channel while that of $^1C_4$ takes noticeably higher values in the midplane than in the

DOI: 10.1021/acs.jpcb.8b05591
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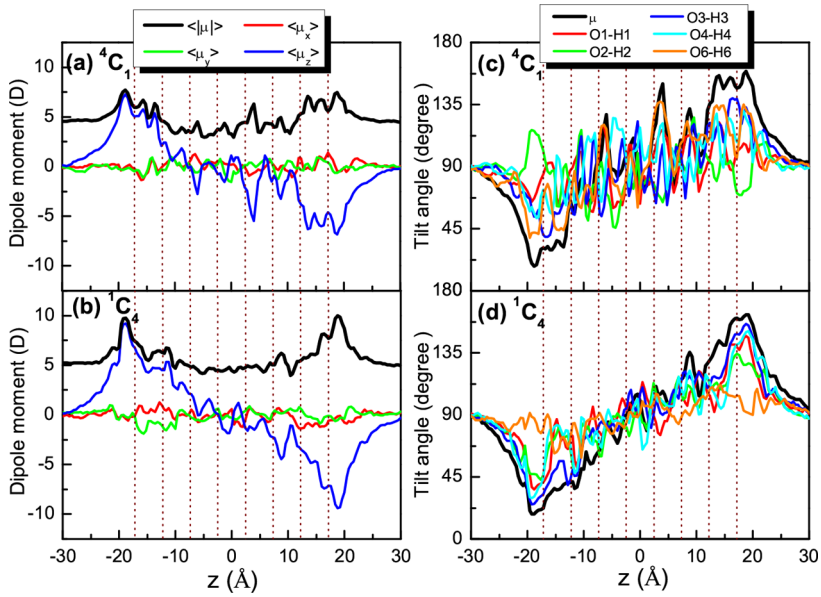


Figure 11. ((a) and (b)) Magnitudes of the dipole moment ($|\mu|$) and average values of each component ($\langle\mu_x\rangle$, $\langle\mu_y\rangle$, and $\langle\mu_z\rangle$) and ((c) and (d)) the tilt angles of the dipole moment and the hydroxy groups of $^4C_1$ and $^1C_4$ as a function of the z coordinate of the CoM of the glucose conformers through the 8 × cyclo[-(Trp-D-Leu)$_4$-Gln-D-Leu-] in DMPC bilayers. The vertical broken lines are the same as those in Figure 2.

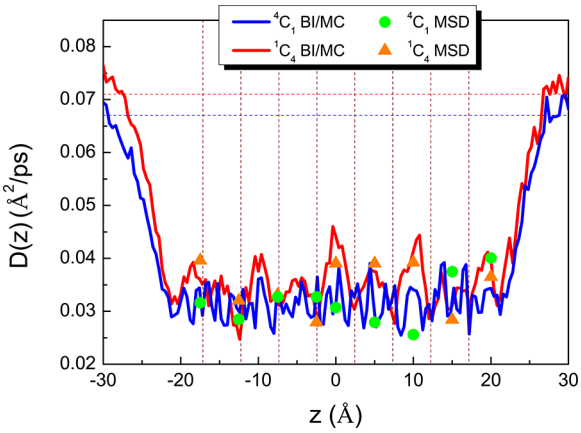


Figure 12. Position-dependent diffusion coefficients of the $^4C_1$ and $^1C_4$ conformers of β-D-glucose through the 8 × cyclo[-(Trp-D-Leu)$_4$-Gln-D-Leu-] in DMPC bilayers calculated using the BI/MC method with the observation time, Δt, of 2.0 ps. Diffusion coefficients of the glucose conformers obtained from the MSD method in several positions inside the CPN are also displayed for comparison. The horizontal broken lines indicate the diffusion coefficients of each conformer in bulk water obtained from MD simulations and the MSD analysis. The vertical broken lines are the same as those in Figure 2.

α-plane regions, indicating that $^1C_4$ has higher translational degrees of freedom at the midplane thanks to its spherical shape and larger volumes of the midplane regions than those of the α-plane regions. The evaluated orientational order parameters for $^1C_4$ in Figure 7 indirectly confirm the higher translational degrees of freedom as well as higher rotational degrees of freedom in the midplane regions compared to those in the α-plane regions.

CONCLUDING REMARKS

In this work, the hydrogen bond network properties, energetics, and transport dynamics for β-D-glucose through a CPN in DMPC bilayers were analyzed using MD simulations and statistical methods, with emphasis on understanding the behavior the two chair conformations, $^4C_1$ and $^1C_4$. New hydrogen bonds between the hydroxyl hydrogens of the glucose conformers and the carbonyl oxygens in the backbone of the CPN are formed inside the CPN, but hydrogen bonds between the hydroxyl oxygens and the amide hydrogens of the CPN backbone are rarely observed. Compared to the number of hydrogen bonds between the glucose conformers and water molecules in bulk water, the total number of hydrogen bonds of the conformers with the CPN backbone and channel water molecules are reduced due to the confinement of the glucose conformers inside the CPN and the failure of hydrogen bonding of the hydroxyl oxygens of glucose with the amide hydrogens of the CPN backbone. However, intramolecular hydrogen bonds of the $^1C_4$ conformer are still maintained through the channel. As for $^1C_4$, the considerably weaker orientational preferences and higher translational degrees of freedom than $^4C_1$ inside the channel can be attributed to its more spherical shape and intramolecular hydrogen bonds, adding entropic contributions to the lowering of the free energy of $^1C_4$. Orientations of the hydroxyl and hydroxymethyl groups of the both glucose conformers display strong correlations with directions of the dipole moment of the conformers along the channel axis and indicate that the hydroxyl and hydroxymethyl groups play a crucial role in electrostatic interactions between a glucose molecule and CPNs as well as hydrogen bond interactions. Our study reveals that transport behavior of glucose through CPNs is strongly influenced by conformational variations and the intramolecular hydrogen bond activity of glucose, providing insight into the biological protein−glucose recognition such as the translocation mechanism of D-glucose into the cytoplasmic region through GLUT proteins and the dynamics of D-glucose near the biological interfaces. It is also demonstrated that CPNs are indeed capable of hosting hydrophilic biomolecules such as glucose owing to the hydrophilic nature of the interior of the CPNs. This indicates that CPNs can be employed as a novel

DOI: 10.1021/acs.jpcb.8b05591
J. Phys. Chem. B 2018, 122, 8174−8184

drug delivery system for hydrophilic drug molecules into the cytoplasmic region of a cell.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcb.8b05591.

One figure (Figure S1) to show the position-dependent diffusion coefficients of $^4C_1$ and $^1C_4$ through the CPN in DMPC bilayers with three different observation times Δt (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This research was supported by Basic Science Research Program (No. NRF-2017R1D1A3B03028669) and by Basic Research Laboratory (No. NRF-2017R1A4A1015405) through the National Research Foundation of Korea (NRF) grant funded by the Korean Government (MEST). This study was supported by 2017 Research Grant from Kangwon National University (No. 520170531). G.C.S. was supported by the Department of Energy, Office of Basic Energy Sciences, under the CBES EFRC, grant DE-SC0000989.

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DOI: 10.1021/acs.jpcb.8b05591
J. Phys. Chem. B 2018, 122, 8174−8184